



Review

Maternal air pollution exposure and congenital heart defects in offspring: A systematic review and meta-analysis



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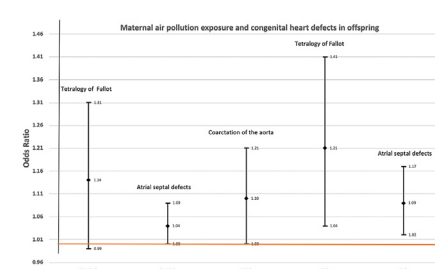
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HIGHLIGHTS

- Whether maternal exposure to air pollution contributes to the development of CHDs in offspring is a controversial question.
- We conducted a systematic review and meta-analysis of epidemiological studies investigating this question.
- Twenty-six studies met our eligibility criteria and were included in our systematic review and meta-analysis.
- Overall, there was notable variability in CHDs definitions, air pollution exposure assessment and confounder adjustment.
- The study suggests significant associations between maternal air pollution exposure and increased risk of specific CHDs subtypes in offspring.

GRAPHICAL ABSTRACT



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ABSTRACT

Background: Congenital heart defects (CHDs) has a multifactorial causation with a strong genetic component and many environmental triggers. Emerging body of empirical studies suggest that air pollution is an important contributor to the development of CHDs, however, there still remains some controversy over the current evidence, and to the authors' knowledge, no studies have reviewed the most recent evidence.

Objectives: We performed a systematic review and meta-analysis of epidemiological literature to investigate the relationship between maternal air pollution exposure and CHDs risk in offspring. The presence of heterogeneity and publication bias across available studies were also examined.

Methods: An extensive literature search of epidemiological studies pertaining to air pollution and CHDs, published in English language, until August 1, 2019 was conducted. Summary risk estimates of pollution

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Meta-analysis
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—outcome combinations were calculated for i) risk per specific increment of concentration and ii) risk at high versus low exposure level in each study using fixed-effect model or random-effects model.

Results: A total of 26 studies were finally included. In the meta-analyses, high versus low carbon monoxide (CO) exposure was associated with an increased risk of tetralogy of Fallot [odds ratio (OR) = 1.21, 95% confidence interval (CI): 1.04–1.41], yet particulate matter $\leq 5 \mu\text{m}$ (PM_{2.5}) exposure was marginally associated with it. Increased risk of atrial septal defects (ASDs) was found for each 10 $\mu\text{g}/\text{m}^3$ and 10 ppb increment in particulate matter $\leq 10 \mu\text{m}$ (PM₁₀) and ozone (O₃) exposure, respectively (OR = 1.04, 95% CI: 1.00–1.09; OR = 1.09, 95% CI: 1.02–1.17). Categorical nitrogen dioxide (NO₂) exposure was associated with an increased risk of coarctation of the aorta (OR for high versus low = 1.14, 95% CI: 1.02–1.26). Analyses for other combinations yielded none statistically significant associations. Sensitive analyses showed similar findings.

Conclusions: The summary effect estimates from this study suggest statistically significant associations between increased risk of specific CHDs subtypes and PM_{2.5}, PM₁₀, NO₂, CO, and O₃ exposures. Further studies, especially conducted in developing countries, with improvements in exposure assessing, outcome harmonizing, and mechanistic understanding are needed to elaborate the suggestive associations.

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1. Introduction

Congenital malformations are the greatest contributors to infant mortality and one of the leading causes of disability globally (Agay-Shay et al., 2013; Boyle et al., 2018; Garne et al., 2012). Congenital heart defects (CHDs), the most common types of congenital malformations, occur in approximately 4–8/1000 live births (Yagel et al., 1997). The etiology of CHDs, although was not fully understood, has been speculated to be multifactorial and prompted sharper focus on environmental factors such as air pollution suggested to play an important role (Ritz, 2010). Potential mechanisms for the teratogenic role of air pollutants on birth defects have been reported, including the inhibition of cell differentiation, proliferation, or migration; decreased or increased cell death; and DNA damage and oxidative stress (Baccarelli and Bollati, 2009; Pedersen

et al., 2009; Perera et al., 2005; Saenen et al., 2017). These mechanisms make it biological plausible to assess the relationship between maternal air pollution exposure and the risk of CHDs in offspring.

Over the past decades, emerging epidemiological studies have examined the role of maternal air pollution exposure in the occurrence of congenital anomalies in offspring, but results were controversial (Agay-Shay et al., 2013; Dadvand et al., 2011a, 2011b; Dolk et al., 2010; Farhi et al., 2014; Gianicolo et al., 2014; Gilboa et al., 2005; Girguis et al., 2016; Hansen et al., 2009; Huang et al., 2019; Hwang et al., 2015; Lavigne et al., 2019; Liu et al., 2017; Padula et al., 2013; Pedersen et al., 2017; Rankin et al., 2009; Ren et al., 2018; Ritz et al., 2002; Salavati et al., 2018; Schembari et al., 2014; Stingone et al., 2017; Strickland et al., 2009; Tanner et al., 2015; Vinceti et al., 2016; Vinikoor-Imler et al., 2013; Zhang et al.,

2016a, 2016b). Two meta-analyses were previously published examining the relationship between maternal exposure to air pollution and risk of congenital anomalies in offspring (Chen et al., 2014; Vrijheid et al., 2011). However, only five defects/defect groupings were explored, let alone the evidence that had been evolving since the last systematic review (Girguis et al., 2016; Huang et al., 2019; Hwang et al., 2015; Jin et al., 2015; Lavigne et al., 2019; Liu et al., 2017; Pedersen et al., 2017; Ren et al., 2018; Salavati et al., 2018; Stingone et al., 2017; Tanner et al., 2015; Vinceti et al., 2016; Zhang et al., 2016a, 2016b). Therefore, we performed a systematic review and meta-analysis of existing literature, including the newly published studies after the last systematic review, to investigate the relationship between maternal air pollution exposure and risk of CHDs in offspring. Unlike previous meta-analyses, we only focused on CHDs specifically in the present study. Meanwhile, we further conducted sensitive analyses to test the robustness of results and also assess the publication bias across available studies.

2. Methods

2.1. Search strategy

This study was performed according to the PRISMA guideline (Moher et al., 2009), attached in the supplementary materials. We conducted an extensive literature, formatting the research question in accordance with the PECOS framework [Population, Exposure, Comparator(s), Outcomes, and Study design] through August 1, 2019 on the following electronic databases: Web of Science, Embase, and PubMed. Table S1 shows the full details of the research question definition and related database search strings. Relevant articles were identified through specific search strategy, with boolean operators AND in combination with search terms for air pollution exposure and CHDs including: “air pollutants”, “air pollution”, “particulate matter”, “nitrogen dioxide”, “nitrogen oxides”, “ozone”, “sulfur dioxide”, “carbon monoxide”, “NO₂”, “NO_x”, “PM_{2.5}”, “PM₁₀”, “SO₂”, “CO”, “O₃”, and “congenital abnormalities”, “congenital anomalies”, “birth defects”, “cardiac anomalies”, “cardiac defects”, “congenital heart diseases”, “congenital heart defects”. We also additionally searched the references of retrieved original studies and reviews concern this topic manually. All titles or abstracts were screened to determine the suitability of initial articles.

2.2. Study selection

Studies met the following criteria were included: i) were published epidemiological studies written in English; ii) explicitly defined subgroup(s) of CHDs as outcome(s); iii) studied maternal air pollution exposure with measured air pollutant concentrations; and iv) provided quantitative effect estimate. Studies used purely ecologic exposure assessments, quantitative traffic density, or proximity to major roads data but without pollutant data were not included in our study. Accordingly, if studies didn't meet the above criteria were excluded.

2.3. Data extraction

A predefined data template was used to extract the following information from eligible studies: study reference (authors and publication year), study setting and time period, exposure measurement, pollutant(s), maternal exposure window(s), exposure distribution, study outcome(s), as well as confounders adjustment in the analysis. Two authors (C.Y.H. and Y.F.) extracted the information using a standard form independently and any discrepancy

was solved by group discussion before the meta-analysis. If we failed to extract necessary information from supplementary materials or associated publications of the selected studies, we would contact the corresponding/first author.

2.4. Quality assessment

We systematically assessed the trustworthiness, relevance, and results of included studies through the checklists developed by the Critical Appraisal Skills Programme (CASP). The CASP checklists are given in form of 12 questions for a cohort study appraisal and 11 questions for a case-control study appraisal, by answering the question in “yes”, “no”, and “can't tell”. Previous systematic review and meta-analysis (Khreis et al., 2017) have summarized the CASP checklists as the following six key parameters: (1) potential for selection bias; (2) outcome measurement or classification bias; (3) exposure measurement, recall, or classification bias; (4) identification of and adjustment for important confounders; (5) length and completion of follow up; and (6) any special characteristics. The summarized checklists were also used in our study because of the similar research topic between present study and the previous one. Meanwhile, we also quantitatively assessed the quality of cohort and case-control studies using the Newcastle-Ottawa scale (NOS) (Yang et al., 2020). A “star system” has been developed in which a study is judged on three broad perspectives: the selection of the study groups; the comparability of the groups; and the ascertainment of either the exposure or outcome of interest for case-control or cohort studies respectively. Each star was assigned to one score and a maximum of 9 score would be assigned to each study and study with a score no less than 7 was considered as “high quality”; otherwise, the study was considered as “low quality”.

2.5. Statistical analysis

Meta-analyses were performed to summarize the risk estimates for the relationship between maternal air pollution exposure and risk of CHDs in offspring. We conducted the meta-analyses if more than four studies reported the risk estimates for the same pollution-outcome combination (Vrijheid et al., 2011). Two types of pooled risk estimates were calculated in our meta-analyses: i) for cardiac defect risk per specific increment of concentration and ii) for the comparison of cardiac defect risk at high vs. low air pollution exposure (e.g., the fourth quartile vs. the first quartile) in individual study. For each of these, we extracted risk estimates from single-pollutant models with fully covariates adjustment rather than multiple-pollutant models in each study, due to the fact that not all the included studies adjusted the other pollutants when multiple air pollutants were analyzed, in spite of considerable collinearity among pollutants from the same sources. If studies reported the risk estimates for more than one exposure windows, the most appropriate one for the development of CHDs and most used in the included studies: 3–8 weeks of gestation (or the closest to it) was selected. Strickland et al. (2009) and Stingone et al. (2014) reported two types of ventricular septal defects (VSD) in their studies, namely perimembranous VSD and muscular VSD. We firstly calculated the combined risk estimate for the two types of VSDs using fixed-effect meta-analysis and then entered the combined risk estimate into meta-analyses. There are two studies, conducted in the same area, covered overlapping time periods, and assessing the relationship of maternal SO₂ exposure with some of the same anomaly groups risk (tetralogy of Fallot, VSDs, coarctation of the aorta), and published in three reports (Dadvand et al., 2011a, 2011b; Rankin et al., 2009). We separately entered the risk estimates extracted from these publications in the meta-analysis of maternal SO₂ exposure one by one, treat the inclusion of (Dadvand

et al., 2011a) study as the main analysis, because this one had also focused on other air pollutants and covered a more recent data, another two studies (Dadvand et al., 2011b; Rankin et al., 2009) were entered as sensitivity analyses.

In the high vs. low air pollution exposure meta-analysis, we extracted the reported risk estimates for the study-specific highest compared with the study-specific lowest exposure category, when they were available. If studies only reported risk estimates for continuous air pollution exposure, the interquartile ranges (IQRs) given in individual studies were used to convert the risk estimate per interquartile increment of each pollutant exposure, because these were thought to be comparable with estimates from the categorical studies (Vrijheid et al., 2011).

For the continuous air pollution exposure meta-analysis, when linear associations were assumed, we extracted the risk estimate per increment in pollutant concentration whenever data is available. The risk estimates were pooled in meta-analyses after standardizing the increment of pollutant concentration as follows: 10 $\mu\text{g}/\text{m}^3$ for $\text{PM}_{2.5}$ and PM_{10} , 10 ppb for O_3 and NO_2 , 1 ppb for SO_2 , and 1 ppm for CO, because this standardization allowed comparability between previous studies (Chen et al., 2014; Vrijheid et al., 2011). If studies reported different expressed metrics of concentration for the same pollutant (e.g., 10 $\mu\text{g}/\text{m}^3$ and 10 ppb for NO_2), we scaled them using different factors and the risk estimates after standardized were shown in Table 2.

The summary risk estimates were calculated using fixed-effect or random-effects meta-analyses. We firstly tested heterogeneity in the risk estimates for each pollution-outcome pair analysis by means of the Q and I^2 statistics. The Cochran's Q statistic is calculated by summing the squares of the deviations of the effect estimates from each study from the overall effect estimate by weighting each study's contribution, and a p value was obtained by comparing the Q statistics with a chi-square distribution with k (number of included studies) minus 1 degrees of freedom. The I^2 statistic describes the percentage of total variation across studies due to heterogeneity rather than chance and is calculated as $(Q - \text{df})/Q \times 100\%$, where Q is the χ^2 test statistic and df denotes the corresponding degrees of freedom. When p value < 0.1 or $I^2 > 50\%$, a significant heterogeneity was defined and we used a random effects model, with the DerSimonian and Laird method. Otherwise, a fixed effect model was performed following the Mantel-Haenszel method (Mantel and Haenszel, 1959). Meta-analyses were conducted to calculate the inverse-variance weighted summary risk estimates and 95% CIs across studies by means of fixed-effect or random-effects models. We further conducted sensitivity analysis by recalculating the pooled risk estimate after omitting one study at a time (leave-one-out method) for the purpose of evaluating its influence on the pooled risk estimate. Publication bias was visually assessed using a funnel plot for asymmetry and Egger's test, a p value < 0.1 indicated as the existence of publication bias. We used the "trim and fill" method to estimate and adjust for the numbers and outcomes of missing studies. All analyses were conducted using the R software (version 3.2.5) and the package "metafor" was used.

3. Results

3.1. Literature selection and study characteristics

A total of 26 studies including 16 case-control and 10 cohort studies were finally included in our study. The detailed study selection process was shown in Fig. 1. Table 1 summarized the detailed characteristics of included studies. Overall, these studies included more than 8 million births worldwide, published from 2002 to 2019, and 13 of them were published after the last

systematic review and meta-analysis (Chen et al., 2014) investigating the relationship between air pollution and congenital anomalies. Dadvand et al. and Zhang et al. divided their studies into two substudies. Among these studies, 8 were conducted in the USA, 4 in England, 4 in China, 2 in Taiwan, 2 in Italy, 2 in Israel, 1 in Spain, 1 in Denmark, 1 in Canada, and 1 in Australia. Almost all the included studies reported at least one of the following outcomes including the overall CHDs, ASDs, VSDs, and tetralogy of Fallot. Most studies evaluated the effect of routinely monitored air pollutants including PM_{10} (17 studies), NO_2 (16 studies), SO_2 (14 studies), O_3 (14 studies), CO (11 studies), and $\text{PM}_{2.5}$ (10 studies). Less frequently studied pollutants were ultrafine particles (UFPs), $\text{PM}_{\text{coarse}}$, $\text{PM}_{2.5}$ absorbance, $\text{PM}_{2.5-10}$, total suspended particulate (TSP), black smoke (BS), nitric oxide (NO), and nitrogen oxides (NO_x). The concentration distributions of these pollutants were varied across the studies (Table 1).

Assessments of air pollution exposure were normally based on the air pollutant measurements that were carried out at fixed site monitoring stations, except Lavigne et al. (2019) who assigned residential exposure to UFPs using mobile monitoring data. Personal exposures were assigned using the monitoring station nearest to the maternal residence at the time of the birth or only included the subjects in different fixed radiuses of monitors, except that Dadvand et al. (2011b) and Schembri et al. (2014) developed a spatiotemporal model in which air pollutant concentrations at the maternal residence were predicted using monitoring data in combination with traffic, meteorology, and land cover data. Geo-statistical interpolation methods including kriging (Farhi et al., 2014; Huang et al., 2019) and inverse distance weighting (Hwang et al., 2015; Liu et al., 2017; Jin et al., 2015) were also used, while Tanner et al. (2015) used inverse-distance weighted spatiotemporal model. Vinceti et al. (2016) and Pedersen et al. (2017) used a dispersion model, Vinikoor-Imler et al. (2013) used a hierarchical Bayesian model that combining modeled air pollution concentration with air monitoring data. Griguis et al. (2016) used satellite remote sensing, meteorological and land use data to assess the air pollution exposure. Particularly, Lavigne et al. (2019) used different methods to reconstruct exposures to the different pollutants of interest. The Georgia study (Strickland et al., 2009) differed from others in that temporal data were derived from one central monitoring station in the county and then were related to the vulnerable window of each pregnancy. Distances from the maternal residence to the monitors, the number and density of monitors are also varied across the included studies. Nearly all the included studies used the established critical exposure window (usually 3–8 weeks of gestation or exposure covers this period) during pregnancy, in which maturing organs or organ systems are vulnerable to teratogens. In the study performed by Dolk et al. (2009), they assigned annual mean pollutant concentrations to the census ward level of residence and were unable to estimate the exposure in pregnancy time windows.

3.2. Quality assessment

To evaluate the risk of bias, we used the Critical Appraisal Skills Programme (CASP) checklists for observational studies. As shown in Table S2, almost all of the included studies were considered as having a good quality to make an appropriate assessment of the association of maternal air pollution exposure with risk of CHDs in offspring, because of the clearly demonstrated aims, important confounders adjustment, opportune measurements of exposure and outcome in the majority of studies. Some of the limitations identified relate to potential risk of misclassification and non-representative samples. Additionally, all of the selected studies were assessed with a minimum of 7 score, and were also considered as high-quality studies according to the NOS (Table S4).

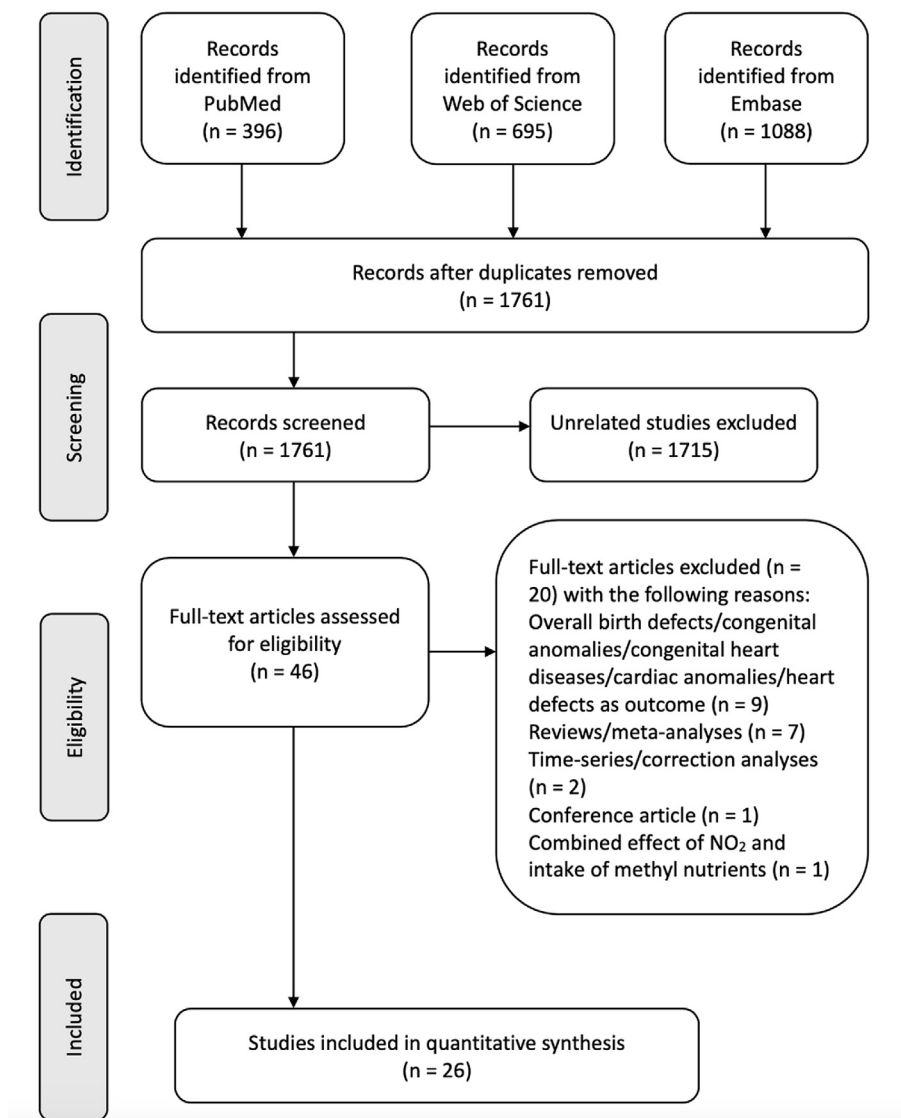


Fig. 1. Flow chart of the study selection process.

3.3. Data synthesis

3.3.1. Overview of meta-analyses

Among the selected studies, each study included as few as 1 and as many as 27 separate CHDs subgroups and investigated these with one to seven pollutant groups. Most studies have reported at least one or a few statistically significant increases in risks with increased air pollution exposure among multiple associations. Various inverse associations were also observed. We conducted meta-analyses for 57 pollutant–cardiac defect combinations for which four or more studies have reported results. The summary risk estimates from these meta-analyses were generally close to one, with a range of summary ORs for continuous exposure from 0.76 to 1.57, and for high vs. low exposure from 0.84 to 1.23. Heterogeneity tests showed evidence of heterogeneity among studies ($P < 0.1$) in less than half of the analyses we conducted. Both results of continuous and categorical exposure meta-analyses are numerically presented in Table 2, along with the heterogeneity parameters, the number of cases, and studies included in each analysis. Overall results for each pollutant are provided as follows.

3.3.2. Meta-analytic summary risk estimates in association with $PM_{2.5}$ exposures

Statistically significant increase in risk of VSDs in relation to $PM_{2.5}$ exposure was observed in two studies (OR per $2.4 \mu\text{g}/\text{m}^3 = 1.18$; 95% CI, 1.01–1.38; OR per $10 \mu\text{g}/\text{m}^3 = 1.16$; 95% CI, 1.03–1.30) (Lavigne et al., 2019; Zhang et al., 2016a). Meta-analyses summarized risk estimates for exposure to $PM_{2.5}$ and risk of ASDs, VSDs, tetralogy of Fallot, and coarctation of the aorta are reported in Fig. 2 and the pooled effect estimates were all close to one, and none reached statistical significance, except an association of borderline statistical significance was observed when comparing the highest vs. lowest exposure categories of $PM_{2.5}$ (OR = 1.14; 95% CI, 0.99–1.31) (Table 2). We found evidence of significant heterogeneity in the analyses of ASDs and VSDs (Table 2).

3.3.3. Meta-analytic summary risk estimates in association with CO exposures

Dadvand et al., 2011a reported statistically significant increase in risk of VSDs (OR per $1 \text{ mg}/\text{m}^3 = 2.63$; 95% CI, 1.87–3.71; OR quartile 4 vs. quartile 1 = 1.66; 95% CI, 1.37–2.02). Ritz et al. (2002) reported that per 1 ppm increment in CO exposure was associated

Table 1
Description of studies included in the systematic review and meta-analysis.

Reference (author, year)	Study setting (time period)	Study design	Exposure measurement	Pollutant(s)	Exposure period	Exposures distribution	Congenital cardiac anomaly associated outcome assessment	Adjustment variables
Ritz et al., (2002)	California, USA (1987–1993); 754,030 births	Case-control	Average of 24-h measurements of pollutants at nearest monitoring station (within 16 km)	NO ₂ , PM ₁₀ , CO, O ₃	Months 1, 2, and 3; trimesters 2 and 3; 3 months preconception	25th–75th percentile: CO (ppm): 1.14–2.39 O ₃ (pphm): 1.07–2.86	Aortic artery and valve defects, pulmonary artery and valve defects, conotruncal defects, aortic defects, pulmonary valve defects, conotruncal defects, VSDs, atrium and ASDs, endocardial and mitral valve defects	Maternal age, race, education, prenatal care, infant sex, decade of birth, parity, birth type, season of conception, and other air pollutants
Gilboa et al., (2005)	Texas, USA (1997–2000); 607,500 births	Case-control	Average hourly or daily concentrations of CO, NO ₂ , O ₃ , PM ₁₀ , and SO ₂ , measured at the nearest, or next nearest, monitor in the county (median distance, 8.6–14.2 km; maximum, 36–54 km)	PM ₁₀ , CO, SO ₂ , NO ₂ , O ₃	3–8 weeks of pregnancy	25th–75th percentile PM ₁₀ (µg/m ³): 19.5–29 CO (ppm): 0.4–0.7 NO ₂ (pphm): 1.3–2.1 O ₃ (pphm): 1.8–3.1 SO ₂ (ppb): 1.3–2.7	Aortic artery and valve defects, ASDs, pulmonary artery and valve defects, VSDs, conotruncal defects, endo-cardial cushion, and mitral valve defects	Maternal age, education, race/ethnicity, season of conception, plurality, parity, infant sex, marital status, prenatal care, illness, gravidity, tobacco use, vital status, year of birth, and county
Dolk et al., (2009)	Wessex, North West Thames, Oxford, and Northern of UK (1991–1999); 759,993 births	Cohort	Annual mean concentrations in 1996 of NO ₂ , PM ₁₀ , and SO ₂ for 1 × 1 km grids; population-weighted average assigned to census ward of residence	SO ₂ , NO ₂ , PM ₁₀	1996 average	10th–90th percentile SO ₂ (µg/m ³): 3.9–25.0 NO ₂ (µg/m ³): 21.5–47.8 PM ₁₀ (µg/m ³): 18.8–26.4	Cardiac defects, anomalies of cardiac chambers, transposition of great vessels, coarctation of aorta, malformations of cardiac septa, atrioventricular septal defects, tetralogy of Fallot, hypoplastic left heart syndrome, malformations of valves, great arteries and veins	Maternal age, area socioeconomic deprivation, and hospital catchment area
Hansen et al., (2009)	Brisbane, Australia (1998–2004); 150,308 births	Case-control	Daily average exposures at 18 monitors with the majority located within a 30 km radius of city	PM ₁₀ , CO, NO ₂ , SO ₂ , O ₃	3–8 weeks of pregnancy	Mean (min, max) PM ₁₀ (µg/m ³): 18.0 (4.4, 151.7) NO ₂ (ppb), 8.2 (1.4, 22.7) O ₃ (ppb), 25.8 (4.3, 54.4) SO ₂ (ppb), 1.5 (0, 7.1) CO (ppm), 1.1 (0.02, 7.0)	Aortic artery and valve defects, ASD, pulmonary artery and valve defects, VSD, conotruncal defects, endocardial cushion and mitral valve defects	Neonate sex, mother's age, marital status, indigenous status, number of previous pregnancies, month of LMP, area-level SES, and distance to pollution monitor
Rankin et al., (2009)	Northern region, UK (1985–1990); 18,197 births	Case-control	Daily average exposures during the first trimester from 62 monitors within 10 km of maternal residence	BS and SO ₂	First trimester	Median (IQR) BS (µg/m ³): 1725 (1593) (controls), 1620 (1555) (cases) SO ₂ (µg/m ³): 3558 (1491) (controls), 3505 (1704) (cases)	CHDs, atrio-ventricular septal defect, tetralogy of Fallot, hypoplastic left heart, PDA, coarctation of aorta, VSD	Birth weight, infant sex, and material deprivation
Strickland et al., (2009)	Atlanta, USA (1986–2003); 715,500 births	Cohort	Average of daily concentration from one central monitoring station	PM ₁₀ , NO ₂ , O ₃ , CO, SO ₂	3–7 weeks of pregnancy	IQR PM ₁₀ (µg/m ³): 14.2 O ₃ (ppb): 29.9 NO ₂ (ppb): 5.7 CO (ppm): 0.3 SO ₂ (ppb): 4.0	ASD (secundum), coarctation of the aorta, hypoplastic left heart syndrome, PDA, pulmonary stenosis (valvar), tetralogy of Fallot, transposition of the great arteries, VSD (muscular), VSD (perimembranous), conotruncal defect, left and right ventricular outflow tract defects	Week of year and cubic spline for day of follow-up
Dadvand et al., (2011a)	Northeast of England (1993–2003); 356,767 births	Case-control	Weekly average of pollutants at the nearest of six monitors to maternal residential location (maximum, 56–83 km; median, 8–12 km)	PM ₁₀ , SO ₂ , O ₃ , NO, CO, NO ₂	3–8 weeks of pregnancy	25th–75th percentile PM ₁₀ (µg/m ³): 20.5–30.2 SO ₂ (µg/m ³):	Congenital malformations of cardiac chambers and connections, cardiac septa, pulmonary and tricuspid valves, aortic and mitral	Year of birth, SES, infant sex, season of conception, and degree of urbanity

Table 1 (continued)

Reference (author, year)	Study setting (time period)	Study design	Exposure measurement	Pollutant(s)	Exposure period	Exposures distribution	Congenital cardiac anomaly associated outcome assessment	Adjustment variables
						6.8–15.0 O ₃ (μg/m ³): 33.2–42.4 NO (μg/m ³): 13.3–32.5 NO ₂ (μg/m ³): 29.2–38.4 CO (mg/m ³): 0.39–0.64	valves, and great arteries and veins, ASD, VSD, tetralogy of Fallot, coarctation of aorta, congenital pulmonary valve stenosis	
Dadvand et al. (2011b)	Northeast of UK (1985–1996); 449,355 births	Case-control	Weekly exposure levels by two stage spatiotemporal modeling at each maternal place of residence using traffic data, land cover data, and other predictors	BS and SO ₂	3–8 weeks of pregnancy	25th–75th percentile BS (μg/m ³): 5.93–14.57 SO ₂ (μg/m ³): 17.6–31.2	Congenital malformations of cardiac chambers and connections, congenital malformations of cardiac septa, congenital malformations of pulmonary and tricuspid valves, VSD, ASD, congenital malformations of aortic and mitral valves, congenital malformations of great arteries and veins, tetralogy of Fallot, congenital pulmonary valve stenosis, coarctation of aorta	Birth year, SES, infant sex, season of conception, and degree of urbanity
Agay-Shay et al. (2013)	Tel-Aviv, Israel (2000–2006); 135,527 births	Cohort	Weekly means of exposures during pregnancy weeks 3–8 according the distance from monitor stations to maternal address	PM _{2.5} , NO ₂ , CO, PM ₁₀ , SO ₂ , O ₃	3–8 weeks of pregnancy	Median (IQR) PM ₁₀ (μg/m ³): 43 (27.1) PM _{2.5} (μg/m ³): 22.1 (13.4) CO (ppm): 0.9 (0.6) NO ₂ (ppb): 23.1 (16.9) SO ₂ (ppb): 2.1 (1.8) O ₃ (ppb): 26.5 (31.3)	Multiple CHDs, isolated atrial and ASDs, isolated VSD, isolated PDA	Multi-births, maternal age, maternal marital status, maternal origin of birth, infant's sex, religion, birth year, season of conception, ward-base SES
Padula et al. (2013)	California, USA (1997–2006); 1,641 births	Case-control	Residence-based assignments around stations, with daily average values during the first two pregnancy months were collected; a maximum interpolation radius of 50 km was used	PM ₁₀ , CO, PM _{2.5} , NO, NO ₂ , O ₃	First two months of pregnancy	Range PM ₁₀ (μg/m ³): 7.90–59.32 PM _{2.5} (μg/m ³): 3.57–66.29 CO (ppm): 0.13–1.37 NO (ppb): 0.69–67.34 NO ₂ (ppb): 2.40–38.94 O ₃ (ppb): 10.49–91.92	CHDs groups (27 subtypes)	Maternal race or ethnicity, education, and vitamin use
Vinikoor-Imler et al. (2013)	Texas, USA (2002–2006); TBDR: 1,422,671 births NBDPS: 812 births	Case-control	Estimated averaged concentration provided by the CMAQ model across weeks 3 through 8 for 12 km × 12 km grid	PM _{2.5} and O ₃	First trimester	Median (IQR) TBDR: [PM _{2.5} (μg/m ³): 12.0 (9.4–14.4), O ₃ (ppb): 40.5 (33.4–46.8)] NBDPS: [PM _{2.5} (μg/m ³): 7.9 (6.5–9.5), O ₃ (ppb): 34.9 (30.0–43.4)]	Conotruncal heart defects, septal heart defects, atrioventricular septal defects, obstructive heart defects, anomalous pulmonary venous return, ventricular outflow tract obstruction	Receiving prenatal care during the first trimester, number of previous live births, maternal age, race/ethnicity, education, and urbanicity (TBDR) Receiving prenatal care during the first trimester, number of previous live births, maternal age, race/ethnicity, and maternal education puls folic acid or multivitamin use, smoking, and alcohol consumption during the month prior to conception through the first trimester in another model (NBDPS)
Farhi et al. (2014)	Israel (1997–2004); 216,730 births	Cohort	Monthly average of exposures measured at 117 monitoring stations by using a GIS and Kriging procedure	SO ₂ , PM ₁₀ , NO _x , O ₃	First trimester, second trimester,	Median (IQR) (first trimester) O ₃ (ppb): 32.3 (8.1)	VSD	Maternal age, maternal ethnicity, maternal country of birth, maternal education

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Table 1 (continued)

Reference (author, year)	Study setting (time period)	Study design	Exposure measurement	Pollutant(s)	Exposure period	Exposures distribution	Congenital cardiac anomaly associated outcome assessment	Adjustment variables
			during the first trimester and entire pregnancy according to maternal residence		entire pregnancy	SO ₂ (ppb): 2.63 (1.25) NO _x (ppb): 19.5 (17.6) PM ₁₀ (μg/m ³): 50.1 (16.4)		plurality, season of birth, and infant gender
Gianicolo et al., (2014)	Brindisi, Italy (2001–2010); 945 births	Case-control	Daily average concentration of pollutants measured by 3 monitoring stations over week 3–8 of pregnancy	SO ₂ and TSP	3–8 weeks of pregnancy	Mean ± SD (μg/m ³): CHDs: [SO ₂ , 2.8 ± 1.2 (control), 2.8 ± 1.4 (case); TSP, 29.8 ± 5.8 (control), 30.3 ± 5.0 (case)] VSDs: [SO ₂ , 2.9 ± 1.2 (control), 3.2 ± 1.5 (case); TSP, 30.2 ± 5.9 (control), 30.7 ± 4.9 (case)]	All CHDs and VSDs.	Infant sex, SES, and year of beginning of pregnancy
Schembari et al., (2014)	Barcelona, Spain (1994–2006); 5,238 births	Case-control	Daily spatiotemporal exposure estimates over weeks 3–8 of pregnancy	NO ₂ , NO _x , PM _{2.5} , PM ₁₀ , PM _{coarse} , PM _{2.5} absorbance	3–8 weeks of pregnancy	Median (IQR) NO ₂ (μg/m ³): 55.7 (12.2) NO _x (μg/m ³): 88.7 (27.1) PM ₁₀ (μg/m ³): 38.7 (3.0) PM _{coarse} (μg/m ³): 21.7 (3.6) PM _{2.5} (μg/m ³): 16.6 (2.6) PM _{2.5} absorbance (10 ⁻⁵ /m): 2.65 (0.73)	CHDs, transposition of great vessels, VSD, ASD, tetragy of Fallot, atrioventricular defect, tricuspid atresia and stenosis, pulmonary valve stenosis, coarctation of aorta	Maternal age, conception season, birth/termination year, and SES
Stingone et al., (2014)	Nine states in the USA (1997–2006); 7,960 births	Case-control	Average of daily maximum concentrations using the closest air monitor within 50 km to their residence during early pregnancy	NO ₂ , CO, O ₃ , PM _{2.5} , PM ₁₀	2–8 weeks of pregnancy	10th–90th percentile NO ₂ (ppb): 18.9–45.5 CO (ppm): 0.58–2.13 O ₃ (ppb): 32.2–51.8 PM _{2.5} (μg/m ³): 7.77–19.7 PM ₁₀ (μg/m ³): 14.9–40.6	Anomalous pulmonary venous return, ASD, AVSD, atrioventricular septal defect, coarctation of the aorta, d-transposition of the great arteries, hypoplastic left heart syndrome, left ventricular outflow tract obstructions, pulmonary valve stenosis, right ventricular outflow tract obstructions, total anomalous pulmonary venous return, tetralogy of Fallot, perimembranous VSDs	Maternal race/ethnicity, age educational attainment, household income, maternal smoking status and alcohol consumption during early pregnancy, nativity, and site specific heart defect ratio
Hwang et al., (2015)	Taiwan (2001–2007); 11,957 births	Case-control	Monthly average of pollutants concentration, measured at 72 monitoring stations in Taiwan; inverse distance-weighted average assigned to each residence	PM ₁₀ , NO ₂ , O ₃ , SO ₂ , CO	First trimester	Median (mean ± SD) O ₃ (ppb): 41.69 (42.71 ± 0.08) CO (ppm): 0.67 (0.70 ± 0.01) NO ₂ (ppb): 22.01 (21.84 ± 0.06) SO ₂ (ppb): 4.14 (4.59 ± 0.02) PM ₁₀ (μg/m ³): 56.57 (60.30 ± 0.18)	VSD, ASD, PDA, endocardial cushion defects, pulmonary artery and valve defects, conotruncal defects including tetralogy of Fallot, transposition of the great arteries, truncus arteriosus communis, double outlet right ventricle, aorticopulmonary window	Maternal age, diabetes mellitus, plurality, gestational age, SES, and season of conception

Cohort

Table 1 (continued)

Reference (author, year)	Study setting (time period)	Study design	Exposure measurement	Pollutant(s)	Exposure period	Exposures distribution	Congenital cardiac anomaly associated outcome assessment	Adjustment variables
Jin et al., (2015)	Lanzhou, China (2010–2012); 8,969 births		Average of daily ambient air pollutants measurements by 4 monitoring stations; inverse distance-weighted average assigned to each residence and work place location	PM ₁₀ , NO ₂ , SO ₂	3–8 weeks of pregnancy the first, second trimester, and entire pregnancy	Mean (IQR) (weeks 3–8) PM ₁₀ (μg/m ³): 137.5 (79.8) NO ₂ (μg/m ³): 44.2 (14.4) SO ₂ (μg/m ³): 57.0 (57.6)	CHDs, congenital malformation of great arteries, congenital malformation of cardiac septa, isolated cases of patent ductus arteriosus	Maternal age, education, income, BMI, disease, folic acid intake, therapeutic drug use, smoking, season of conception, fuel used for cooking, and temperature
Tanner et al., (2015)	Florida, USA (2000–2009); 2,648,533 births	Cohort	Daily average of pollutants, measured at 91 PM _{2.5} and 41 benzene stations, inverse distance weighted spatiotemporal average assigned to each mother's geocoded residential address at the delivery time	Benzene and PM _{2.5}	3–4, 3–8, 3–12, and 5–12 weeks of gestation	Median (IQR) Benzene (log10 ppb) and PM _{2.5} (log10 mg/m ³): 0.15 (0.25), 0.95 (0.11) (No malformation); 0.15 (0.25), 0.95 (0.11) (CCHD); 0.16 (0.25), 0.95 (0.11) (Orofacial clefts); 0.14 (0.27), 0.95 (0.16) (Spina bifida)	Dextro-transposition of the great arteries, double-outlet right ventricle, hypoplastic left heart syndrome, tetralogy of Fallot, truncus arteriosus, total anomalous pulmonary venous return, pulmonary atresia, tricuspid atresia, ebstein anomaly, coarctation of the aorta, aortic interruption/atresia/hypoplasia, single ventricle, any critical CHDs	Maternal race/ethnicity, maternal nativity, maternal smoking, maternal education, maternal age group, marital status, neighbourhood median income, parity, and infant sex
Girguis et al., (2016)	Massachusetts, USA (2001–2008) 11,713 births	Case-control	Daily average PM _{2.5} measured at the 4 km grid cell of infant's birth address using satellite remote sensing, meteorological and land use data	PM _{2.5} and traffic-related exposures (distance to roads and traffic density)	3–7, 1–4, and 6–12 weeks of pregnancy	NA	Transposition of the great vessel, tetralogy of Fallot, ostium secundum ASD, endocardial cushion defect, pulmonary valve atresia or stenosis, aortic valve stenosis, hypoplastic left heart syndrome, PDA, coarctation of aorta, pulmonary artery anomalies, VSD, single common atrium, ASD, patent foramen ovale, insufficiency of aortic valve	Plurality, maternal race or ethnicity, education, language preference, delivery payment source, smoking and alcohol consumption during pregnancy, adequacy of prenatal care, marital status, maternal age, birth year, parity, and season of conception
Vinceti et al., (2016)	Reggio Emilia, Italy (1998–2006); 456 births	Case-control	Average of hourly maximum concentration of pollutants, estimated at the geocoded maternal residence of interest through a dispersion model	Benzene and PM ₁₀	First trimester	Mean ± SD PM ₁₀ (μg/m ³): 5.68 ± 2.48 (cases), 5.46 ± 2.19 (controls) Benzene (μg/m ³): 0.62 ± 0.35 (cases), 0.63 ± 0.35 (controls)	Pulmonary valve stenosis, VSD, ASD	Other pollutants, maternal age, hospital, and year of delivery
Zhang et al. (2016a), 2016b	Wuhan, China (2011–2013) 105,998 births	Cohort	Average of daily concentration of pollutants measured at nearest monitoring station to the living residence of mothers during their early pregnancy	CO, SO ₂ , NO ₂ , O ₃	First trimester	Mean (IQR) O ₃ (μg/m ³): 63.6 (74.1) SO ₂ (μg/m ³): 32.6 (35.9) NO ₂ (μg/m ³): 55.8 (39) CO (mg/m ³): 1 (0.59)	All CHDs, VSD, tetralogy of Fallot	Maternal age, education, parity, infant sex, and season of conception
Zhang et al. (2016a), 2016b	Wuhan, China (2011–2013) 105,988 births	Cohort	Average of daily concentration of pollutants measured at nearest monitoring station to the living residence of mothers during their early pregnancy	PM _{2.5} and PM ₁₀	First trimester	Mean (IQR) PM _{2.5} (μg/m ³): 65.61 (47.24) PM ₁₀ (μg/m ³): 101.73 (74.83)	All CHDs, VSD, tetralogy of Fallot	Maternal age, education, parity, infant sex, and season of conception
Liu et al., (2017)	Fuzhou, China (2007–2013); 4,634 births	Case-control	Daily, monthly, and seasonal average measurements of PM ₁₀ by 3 monitoring stations in the built-up region; inverse distance-	PM ₁₀	First two months	Daily mean PM ₁₀ (μg/m ³): 69.4 Monthly mean range	VSD, ASD, PDA, tetralogy of Fallot, pulmonary valve stenosis, transposition of conducting arteries, coarctation of aorta,	Maternal age, parity, educational attainment, prenatal care, prenatal folic acid and vitamin use, season of conception, and marital status

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Table 1 (continued)

Reference (author, year)	Study setting (time period)	Study design	Exposure measurement	Pollutant(s)	Exposure period	Exposures distribution	Congenital cardiac anomaly associated outcome assessment	Adjustment variables
Pedersen et al. (2017)	Denmark (1997–2002) 84,218 liveborn singletons	Cohort	weighted average assigned to each residence Average of daily NO ₂ concentration calculated using the advanced AirGIS dispersion model	NO ₂ , Road traffic noise	First trimester	PM ₁₀ (μg/m ³): 56.1–72.9 Mean (IQR) NO ₂ (μg/m ³): 11.3 (6.6) L _{den} (dB): 57.4 (9.5)	hypoplastic left heart syndrome, others, total CHDs Overall CHDs, ASDs, VSDs, coarctation of aorta, pulmonary valve stenosis, tetralogy of Fallot, severe CHD	Parental age, maternal smoking, maternal alcohol consumption, maternal education, and disposable income
Huang et al., (2019)	Taiwan (2007–2014); 5,474 births	Case-control	Average of daily ambient air pollutants measurements by 66 fix-site monitoring stations, interpolating the daily average concentrations to the level of the township using ArcGIS	PM _{2.5} , O ₃ , PM ₁₀ , CO, SO ₂ , NO ₂ , PM _{2.5-10}	3–8 weeks of pregnancy	Median (IQR) PM ₁₀ (μg/m ³): 49.4 (23.2) PM _{2.5} (μg/m ³): 28.5 (13.4) PM _{2.5-10} (μg/m ³): 20.8 (9.4) NO ₂ (ppb): 17.8 (5.8) SO ₂ (ppb): 3.51 (1.11) CO (ppm): 0.504 (0.174) O ₃ (ppb): 27.7 (7.9)	Major CHDs, as well as its six major subtypes namely VSD, ASD, tetralogy of Fallot, endocardial cushion defect, transposition of the great arteries, and pulmonary artery and valve stenosis	Infant sex, maternal age, season of conception, maternal diabetes and hypertension, maternal smoking during pregnancy, socioeconomic status of the residential township, and birth year
Lavigne et al., (2019)	Toronto, Canada (2006–2012); 158,743 births	Cohort	Weekly air pollutants concentrations derived from different methods assigned to the geographical coordinates representing the centroid of each mother's residential 6- digit postal codes of residence(s)	UFPs, PM _{2.5} , NO ₂ , O ₃	2–8 weeks of pregnancy	Median (IQR) UFPs (count/cm ³): 26,789 (10,864) PM _{2.5} (μg/m ³): 9.1 (2.4) NO ₂ (ppb): 15.3 (7.4) O ₃ (ppb): 24.2 (10.8)	Overall CHDs, ASD, and VSD	Maternal age, infant sex, parity, maternal smoking status during pregnancy, gestational age, birth weight, maternal diabetes, maternal preeclampsia, three neighbourhood-level (SES) variables, random-effects for neighbourhood in the city of Toronto, and random effects for clustering within families

Abbreviations: ASDs, atrial septal defects; VSDs, ventricular septal defects; LMP, last menstrual period; SES, socioeconomic status; BS, black smoke; IQR, interquartile range; CHDs, congenital heart defects; PDA, patent ductus arteriosus; CMAQ, community multiscale air quality; TBDR, Texas Birth Defects Registry; NBDPS, National Birth Defects Prevention Study; TSP, total suspended particulate; SD, standard deviation; AVSD, atrioventricular septal defects; BMI, body mass index; UFPs, ultrafine particles; NA, not available.

with increased risk of VSDs (OR = 1.33; 95% CI, 1.00–1.78). Meta-analyses summarized risk estimates for exposure to CO and risk of ASDs, VSDs, conotruncal defects, and tetralogy of Fallot are reported in Fig. 3. High vs. low comparisons of CO exposure showed evidence for an increased risk of tetralogy of Fallot (OR = 1.21; 95% CI, 1.04–1.41). Evidence of heterogeneity was observed for the analyses of VSDs and conotruncal defects (Table 2).

3.3.4. Meta-analytic summary risk estimates in association with O₃ exposures

Lavigne et al. (2019) and Gilboa et al. (2005) reported that O₃ exposure was associated with increased risk of ASDs (OR per 10.8 ppb = 1.22; 95% CI = 1.02–1.45; OR per 10 ppb = 1.27; 95% CI = 1.01–1.61). Hwang et al. (2015) reported that O₃ exposure increased the risk of VSDs (OR per 10 ppb = 1.31; 95% CI, 1.10–1.57; OR quartile 4 vs. quartile 1 = 2.34; 95% CI, 1.41–3.90), ASDs (OR quartile 4 vs. quartile 1 = 1.87; 95% CI, 1.14–3.09), patent ductus arteriosus (OR per 10 ppb = 1.19; 95% CI, 1.01–1.40). Zhang et al. (2016b) reported that per 10 μg/m³ increment of O₃ exposure was associated with increased risk of tetralogy of Fallot (OR = 1.24; 95% CI, 1.07–1.44). Meta-analyses summarized risk estimates for exposure to O₃ and risk of ASDs, VSDs, conotruncal defects, tetralogy of Fallot, coarctation of the aorta, and pulmonary valve stenosis are reported in Fig. 4, only increased risk of ASDs was found (OR per 10 ppb = 1.09; 95% CI, 1.02–1.17), while the others did not reach statistical significance. Heterogeneity tests showed evidence for

heterogeneity only related to analyses of VSDs (Table 2).

3.3.5. Meta-analytic summary risk estimates in association with PM₁₀ exposures

Two studies (Gilboa et al., 2005; Strickland et al., 2009) reported statistically significant increases in risk of ASDs and patent ductus arteriosus, respectively (OR per 10 μg/m³ = 1.33; 95% CI, 1.11–1.60; OR per 14.2 μg/m³ = 1.60; 95% CI, 1.11–2.31). Meta-analyses summarized risk estimates for exposure to PM₁₀ and risk of ASDs, VSDs, coarctation of the aorta, tetralogy of Fallot, and patent ductus arteriosus are reported in Fig. 5. We also found a statistically significant increase in risk of ASDs for PM₁₀ exposure (OR per 10 μg/m³ = 1.04; 95% CI, 1.00–1.09). Furthermore, significant heterogeneity was observed in the analyses of the associations between categorical PM₁₀ exposure and ASDs, continuous and categorical PM₁₀ exposures and patent ductus arteriosus.

3.3.6. Meta-analytic summary risk estimates in association with NO₂ exposures

Schembari et al. (2014) reported that per 12.2 μg/m³ increment of NO₂ exposure was associated with increased risk of coarctation of aorta (OR = 1.15; 95% CI, 1.01–1.31). Meta-analyses summarized risk estimates for exposure to NO₂ and risk of ASDs, VSDs, coarctation of the aorta, tetralogy of Fallot, patent ductus arteriosus, and pulmonary valve stenosis are reported in Fig. 6, and only (borderline) statistically significant increases in risk of coarctation of the

Table 2

Summary of meta-analysis of studies on air pollutant exposures and cardiac anomalies.

Pollutant and CHD combination	Studies Included ^a	Total number of cases ^b	Continuous exposure ^c			High versus low exposure		
			Heterogeneity <i>P</i> -value, <i>I</i> ² (%)	Summary OR (95% CI) ^d	Egger's test <i>P</i> -value	Heterogeneity <i>P</i> -value, <i>I</i> ² (%)	Summary OR (95% CI) ^d	Egger's test <i>P</i> -value
PM _{2.5} per 10 µg/m ³								
ASDs	9, 10, 14, 15, 19, 25, 26	2,184	0.02, 65	1.08 (0.87, 1.34)	0.509	<0.01, 79	0.91 (0.74, 1.12)	0.182
VSDs	9, 10, 14, 15, 19, 21, 25, 26	2,070	<0.01, 75	1.04 (0.87, 1.25)	0.852	<0.01, 77	0.99 (0.80, 1.21)	0.877
Tetralogy of Fallot	10, 14, 15, 18, 19, 21, 25, 26	375	0.99, 0	1.12 (0.98, 1.28)	0.983	0.85, 0	1.14 (0.99, 1.31)	0.014
Coarctation of the aorta	10, 15, 18, 26	853	NA			0.91, 0	1.02 (0.87, 1.19)	0.095
CO per 1 ppm								
ASDs	1, 2, 4, 6, 8, 9, 15, 16, 25	2,220	0.15, 35	0.85 (0.72, 1.01)	0.023	0.12, 37	0.86 (0.77, 0.96)	0.069
VSDs	1, 2, 4, 6, 8, 9, 10, 15, 16, 21, 25	4,652	<0.01, 83	1.04 (0.69, 1.56)	0.537	<0.01, 84	1.00 (0.80, 1.24)	0.821
Conotruncal defects	1, 2, 4, 6, 16	1,133	<0.01, 77	0.76 (0.44, 1.33)	0.063	<0.01, 78	0.84 (0.56, 1.25)	0.393
Tetralogy of Fallot	2, 6, 8, 13, 15, 16, 21, 25	688	0.93, 0	1.57 (0.93, 2.64)	0.829	0.49, 0	1.21 (1.04, 1.41)	0.856
O ₃ per 10 ppb								
ASDs	1, 2, 4, 6, 8, 9, 10, 15, 16, 25, 26	2,813	0.35, 10	1.09 (1.02, 1.17)	0.532	0.09, 38	1.06 (0.96, 1.18)	0.845
VSDs	1, 2, 4, 6, 8, 9, 10, 12, 15, 16, 22, 25, 26	6,097	<0.01, 70	1.03 (0.91, 1.15)	0.485	<0.01, 62	1.05 (0.92, 1.20)	0.251
Conotruncal defects	1, 2, 4, 6, 11, 16	2,947	0.65, 0	1.03 (0.96, 1.10)	0.241	0.54, 0	1.02 (0.91, 1.14)	0.031
Tetralogy of Fallot	6, 8, 16, 22, 25, 26	741	0.25, 25	1.08 (0.96, 1.22)	0.984	0.20, 32	1.11 (0.90, 1.37)	0.224
Coarctation of the aorta	8, 10, 15, 26	476	NA			0.60, 0	1.08 (0.82, 1.43)	0.822
Pulmonary valve stenosis	8, 10, 15, 26	714	NA			0.58, 0	1.23 (0.94, 1.61)	0.628
PM ₁₀ per 10 µg/m ³								
ASDs	2, 4, 6, 8, 9, 10, 14, 15, 16, 23, 25	1,902	0.09, 43	1.04 (1.00, 1.09)	0.459	<0.01, 72	1.11 (0.90, 1.37)	0.063
VSDs	2, 4, 6, 8, 9, 10, 12, 14, 15, 16, 20, 21, 23, 25	5,638	0.07, 43	1.00 (0.97, 1.03)	0.152	0.63, 0	0.94 (0.89, 0.99)	0.032
Coarctation of the aorta	2, 3, 6, 8, 10, 14, 15	802	0.72, 0	0.98 (0.86, 1.13)	0.436	0.51, 0	1.03 (0.90, 1.17)	0.411
Tetralogy of Fallot	2, 3, 6, 8, 10, 14, 15, 16, 21, 23, 25	983	0.28, 19	1.01 (0.94, 1.09)	0.420	0.22, 23	0.97 (0.86, 1.10)	0.793
Patent ductus arteriosus	6, 9, 16, 17, 23	625	<0.01, 81	1.06 (0.95, 1.19)	0.049	<0.01, 73	1.22 (0.87, 1.71)	0.062
NO ₂ per 10 ppb								
ASDs	2, 4, 6, 8, 9, 10, 14, 15, 16, 25, 24, 26	2,871	0.44, 0	0.98 (0.90, 1.06)	0.589	0.37, 8	0.98 (0.91, 1.04)	0.432
VSDs	2, 4, 6, 8, 9, 10, 14, 15, 16, 22, 25, 24, 26	5,397	0.62, 0	0.97 (0.91, 1.04)	0.199	0.25, 20	0.98 (0.93, 1.04)	0.070
Coarctation of the aorta	2, 3, 6, 8, 10, 14, 15, 24, 26	927	0.57, 0	1.12 (0.99, 1.28)	0.086	0.08, 43	1.10 (1.00, 1.21)	0.736
Tetralogy of Fallot	2, 3, 6, 8, 10, 14, 15, 16, 22, 24, 26	969	0.52, 0	1.11 (0.95, 1.30)	0.938	0.19, 26	1.08 (0.95, 1.23)	0.996
Patent ductus arteriosus	6, 9, 16, 17	625	0.47, 0	1.10 (0.92, 1.31)	0.106	0.46, 0	1.14 (0.93, 1.39)	0.437
Pulmonary valve stenosis	6, 8, 10, 14, 24, 26	679	0.99, 0	0.91 (0.72, 1.15)	0.608	0.80, 0	0.93 (0.80, 1.08)	0.208
SO ₂ per 1 ppb								
ASDs	2, 4, 6, 8, 9, 15, 16, 25	1,792	<0.01, 74	0.94 (0.86, 1.03)	0.537	<0.01, 78	0.97 (0.74, 1.26)	0.558
ASDs	2, 4, 6, 7 ^e , 9, 15, 16, 25	1,789	<0.01, 82	0.95 (0.87, 1.03)	0.456	<0.01, 77	1.00 (0.77, 1.29)	0.665
VSDs	2, 4, 6, 8, 9, 12, 13, 15, 16, 22, 25	5,509	<0.01, 84	0.97 (0.91, 1.03)	0.633	<0.01, 81	0.94 (0.78, 1.14)	0.915
VSDs	2, 4, 6, 7 ^e , 9, 12, 13, 15, 16, 22, 25	5,257	<0.01, 79	0.98 (0.92, 1.05)	0.655	<0.01, 71	0.99 (0.86, 1.15)	0.640
VSD	2, 4, 5 ^f , 6, 9, 12, 15, 16, 22, 25	4,730	NA			<0.01, 72	0.98 (0.84, 1.15)	0.771
Coarctation of the aorta	2, 3, 6, 8, 15	685	0.09, 53	1.01 (0.96, 1.06)	0.829	0.28, 21	1.08 (0.87, 1.33)	0.630
Coarctation of the aorta	2, 3, 6, 7 ^e , 15	687	0.03, 66	1.00 (0.95, 1.06)	0.882	0.01, 70	0.97 (0.68, 1.38)	0.615
	2, 3, 5, 6 ^f , 15	824	NA			0.84, 0		0.669

(continued on next page)

Table 2 (continued)

Pollutant and CHD combination	Studies Included ^a	Total number of cases ^b	Continuous exposure ^c			High versus low exposure		
			Heterogeneity P-value, I ² (%)	Summary OR (95% CI) ^d	Egger's test P-value	Heterogeneity P-value, I ² (%)	Summary OR (95% CI) ^d	Egger's test P-value
Coarctation of the aorta							1.12 (0.95, 1.33)	
Tetralogy of Fallot	2, 3, 6, 8, 15, 16, 22, 25	919	<0.01, 69	0.99 (0.93, 1.06)	0.166	<0.01, 70	0.95 (0.76, 1.20)	0.508
Tetralogy of Fallot	2, 3, 6, 7 ^e , 15, 16, 22, 25	933	0.09, 46	1.01 (0.97, 1.06)	0.475	0.24, 24	1.02 (0.90, 1.17)	0.955
Tetralogy of Fallot	2, 3, 5 ^e , 6, 15, 16, 22, 25	839	NA			0.34, 12	1.04 (0.93, 1.16)	0.883
Patent ductus arteriosus	6, 9, 16, 17	645	0.26, 26	0.98 (0.95, 1.02)	0.447	0.09, 53	0.93 (0.67, 1.30)	0.551

^a Studies included are different in each pollutant–cardiac anomaly meta-analysis depending on the data they published. References are as follows: 1, Ritz et al. (2002); 2, Gilboa et al. (2005); 3, Dolk et al. (2010); 4, Hansen et al. (2009); 5, Rankin et al. (2009); 6, Strickland et al. (2009); 7, Dadvand et al. (2011b); 8, Dadvand et al. (2011a); 9, Agay-Shay et al. (2013); 10, Padula et al., 2013; 11, Vinikoor-Imler et al. (2013); 12, Farhi et al. (2014); 13, Gianicolo et al. (2014); 14, Schembari et al. (2014); 15, Stingone et al. (2014); 16, Hwang et al. (2015); 17, Jin et al. (2015); 18, Tanner et al. (2015); 19, Girguis et al. (2016); 20, Vinceti et al. (2016); 21, Zhang et al. (2016a); 22, Zhang et al. (2016b); 23, Liu et al. (2017); 24, Pedersen et al. (2017); 25, Huang et al. (2019); 26, Lavigne et al. (2019).

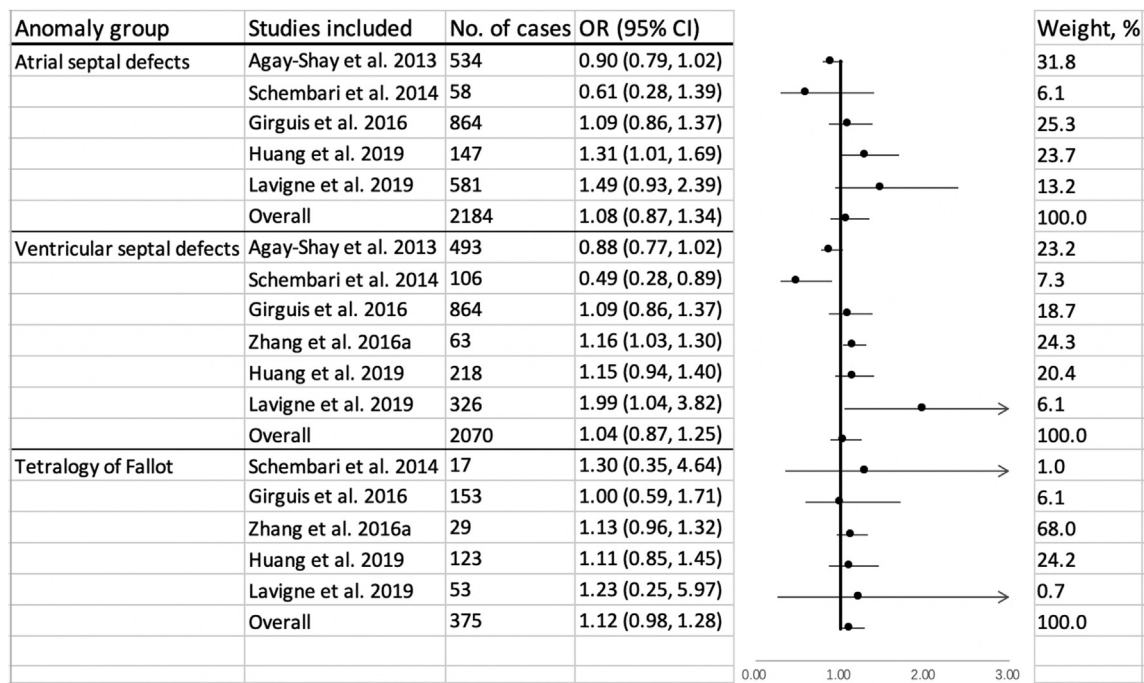
^b Number of cases included in the continuous exposure analysis when both exposures were reported or cases included in the categorical analysis when only categorical analysis available.

^c Conversion factors: CO, 1 ppb = 1.15 µg/m³; NO₂, 1 ppb = 1.88 µg/m³; O₃, 1 ppb = 1.96 µg/m³; SO₂, 1 ppb = 2.62 µg/m³.

^d When heterogeneity p-value is < 0.10, the OR from random effect model is shown; otherwise, the OR from the fixed-effects model is shown.

^e Sensitivity analysis using Dadvand et al. (2011b) instead of Dadvand et al. (2011a).

^f Sensitivity analysis including Rankin et al. (2009) instead of Dadvand et al. (2011a). Summary effect estimates are in bold when the 95% CI do not include 1 and in bold italics for borderline CI values (1.00–...).

Fig. 2. Summary of meta-analysis of studies for the association between PM_{2.5} exposure and congenital heart defects.

aorta were reported (OR per 10 ppb = 1.12; 95% CI, 0.99–1.28; OR high vs. low = 1.10; 95% CI, 1.00–1.21). No evidence of heterogeneity was found for all the NO₂–cardiac defect pairs analyses (Table 2).

3.3.7. Meta-analytic summary risk estimates in association with SO₂ exposures

Statistically significant increases in risk of tetralogy of Fallot and VSDs were reported by Dolk et al. (2009) (OR per 1 µg/m³ = 1.38; 95% CI, 1.07–1.79) and Gilboa et al. (2005) (OR per 1 ppb = 1.14; 95% CI, 1.08–1.21), respectively. Meta-analyses summarized risk

estimates for exposure to SO₂ and risk of ASDs, VSDs, coarctation of the aorta, tetralogy of Fallot, and patent ductus arteriosus are reported in Fig. 7, but none reached statistical significance. Heterogeneity tests showed evidence for heterogeneity among most SO₂–cardiac defect pairs analyses (Table 2).

3.3.8. Sensitivity analyses and publication bias

We re-ran all analyses after systematically excluding each study in turn from the meta-analysis to test the influence of single study (Tables S5–S10). Most of these analyses didn't appreciably change the various summary estimates. But results for the pairs of

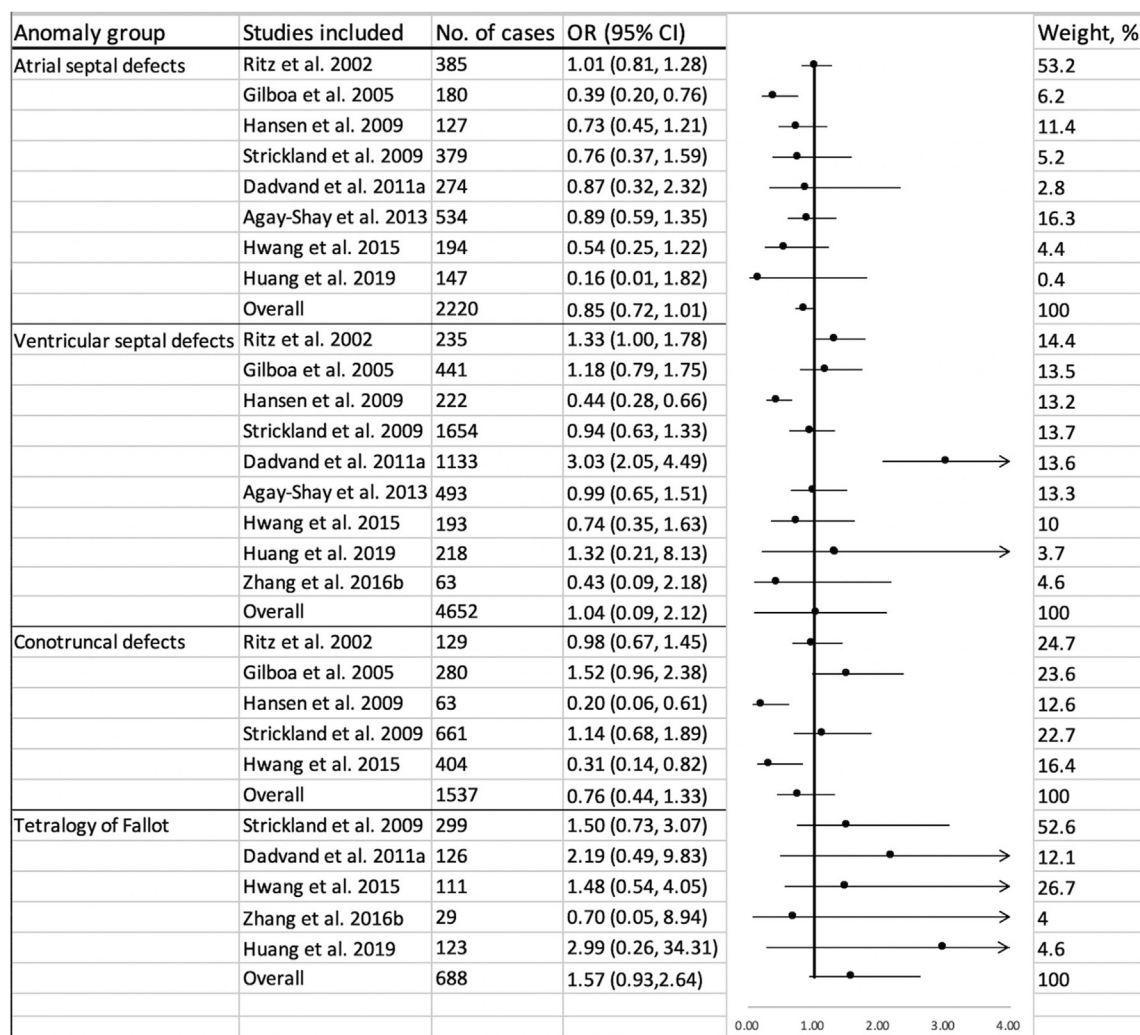


Fig. 3. Summary of meta-analysis of studies for the association between CO exposure and congenital heart disease.

NO₂–coarctation of the aorta and PM₁₀–patent ductus arteriosus were not robust, removing some studies one by one change the significance of summary estimates. Meanwhile, the results for SO₂ were also robust when use of alternative estimates from the first Dadvand study instead of the second (Table 1). Funnel plots for the pollution–cardiac defect combinations are shown in Figs. S1–S57, with slightly asymmetric distribution for few of pollution–cardiac defect combinations, in spite of the fact that there are not enough studies to comprehensively examine publication bias in the present study when using funnel plots. According to the P values of Egger's tests, statistically significant publication biases were found in the case of the categorical PM_{2.5}–tetralogy of Fallot, continuous CO–ASDs, categorical CO–ASDs, continuous CO–conotruncal defects, categorical O₃–conotruncal defects, categorical PM₁₀–ASDs, categorical PM₁₀–VSDs, continuous PM₁₀–patent ductus arteriosus, categorical PM₁₀–patent ductus arteriosus, categorical NO₂–VSDs, and continuous NO₂–coarctation of the aorta combinations (Table 2). After augmenting the observed data with “trim and fill” methods, the pooled risk estimates remained quantitatively almost unchanged. Thus, wide-scale publication bias is unlikely to occur and the effects of publication bias on the pooled risk estimates were negligible.

4. Discussion

To the authors' knowledge, the present study is the largest and most up-to-date systematic review and meta-analysis of current evidence regarding the association of maternal air pollution exposure with CHDs in offspring. Our study suggests statistically significant association between the two, which are also consistent with most of the previous individual studies. Evidence from other syntheses also provided considerable support for the hypothesis that maternal exposure to air pollution plays an important role in the occurrence of CHDs in offspring (Chen et al., 2014; Vrijheid et al., 2011). Meta-analyses detailly suggest that CO exposure was associated with statistically significantly increased risk of tetralogy of Fallot, PM₁₀ and O₃ exposures were related to increased risk of ASDs, while NO₂ exposure was associated with increased risk of coarctation of the aorta. Borderline significant associations were found among the combinations of categorical PM_{2.5}–tetralogy of Fallot and continuous NO₂–coarctation of aorta. Analyses for other pollution–cardiac defect combinations yielded none statistically significantly associations.

Almost all the selected studies using routinely fixed-site air monitored data as measurement basis for maternal personal exposure evaluation, except Lavigne et al. (2019) using mobile monitoring data for UFPs exposure assessment. Exposure indices

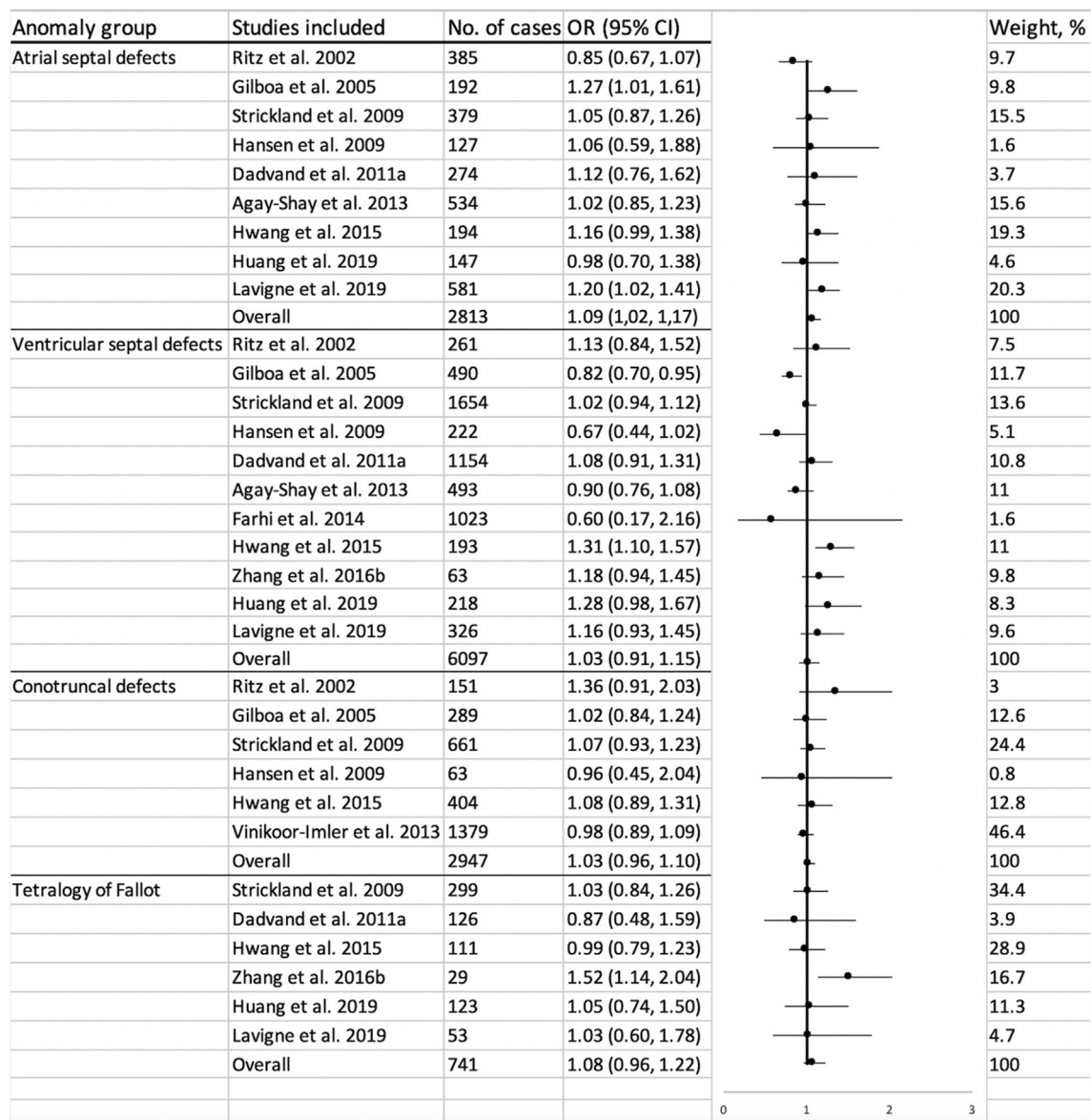


Fig. 4. Summary of meta-analysis of studies for the association between O_3 exposure and congenital heart defects.

were always calculated from pollutant measurements at subjectively chosen cut-off point distance between maternal residence and the nearest monitoring station or as a distance-weighted average of measurements of all stations in study area, to ensure the monitors measured levels were more representative of ambient levels at residence. These studies have largely used measured pollutant concentrations at fixed-site stations to represent the exposure of large populations which didn't consider the widely spatial distributions of air pollutants and may lead to exposure inaccuracies. Although lack of consensus on which air pollution exposure estimate methods are most appropriate presently, more precise spatiotemporal air pollution models including land-use regression models and air dispersion models that take into account the information on land use, population density, traffic volume, distance to pollutant source, and meteorological conditions, are therefore increasingly recommended and used to reduce exposure misclassification in the research field of birth outcomes. Only recently published studies have used these more precise spatial models. Using these exposure assessment modeling would

make more accurate estimate of personal air pollution exposure in future studies regarding the association of air pollution with congenital anomalies. Moreover, some studies (Basu et al., 2014; Hao et al., 2016; Pedersen et al., 2016) assessing the adverse effect of $PM_{2.5}$ on other birth outcomes have moved forward to estimate the effect of specific chemical constituent responsible and future studies can focus on this valuable research topic.

To our knowledge, almost all the included studies were based on birth registries, air pollution exposure estimation was based on the maternal residential addresses at the time of delivery and treated as surrogate for personal exposure, in spite of some studies (Chen et al., 2010; Lupo et al., 2010) have reported that maternal mobility during pregnancy seems doesn't lead to significant change in the assigned maternal air pollution exposure. Furthermore, personal air pollution exposure was not only based the exposure near the home, but also the exposure in different micro-environments including work place, during commutes, and indoor, which are thought to play an important role (Nethery et al., 2008; Setton et al., 2011). However, it's hard to move away from

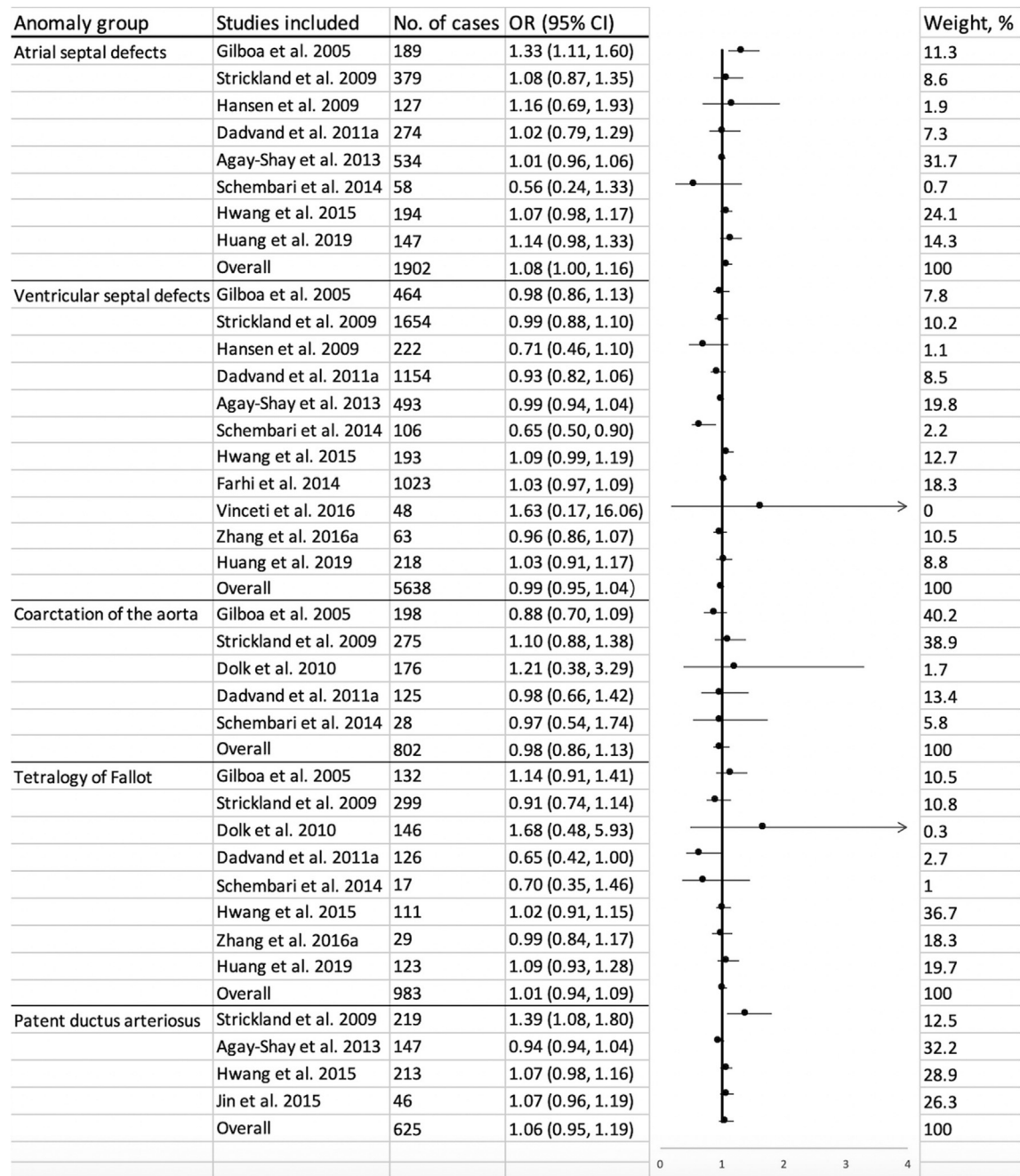


Fig. 5. Summary of meta-analysis of studies for the association between PM_{10} exposure and congenital heart defects.

routinely registration data for relevant research in this field. Large case-control studies with good information on maternal residential mobility, maternal time-activity patterns, and micro-environmental pollutant levels may be a useful step forward, if all the information can be incorporated to accurate personal exposure assessments.

Combining data extracted from the studies investigating the relationship between maternal air pollution exposure and CHDs risk in offspring was difficult because of the differences in the definition of CHDs, classification of CHDs subgroups, and inclusion/exclusion criteria used in the selected studies. In the present study, we only included a few relatively comparable CHDs subgroups for risk estimation synthesis. However, even for these, individual

studies used different classification approaches. For example, VSDs were one of the major subgroups of CHDs and were treated as one anomaly group by most included studies, as different anomalies (VSDs—perimembranous, VSDs—muscular, VSDs—cono) by other studies (Girguis et al., 2016; Padula et al., 2013; Stingone et al., 2014; Strickland et al., 2009), and were excluded in yet another study (Dolk et al., 2009). Consequently, the largest evidence for heterogeneity was found for most air pollutant–VSDs combinations. Generally speaking, it's notoriously difficult to classify as CHDs form a very heterogeneous set of conditions (Botto et al., 2007; Hansen et al., 2009), and because of the varied diagnosis level among different hospitals, which could also introduce heterogeneity. Although, several classification systems have been

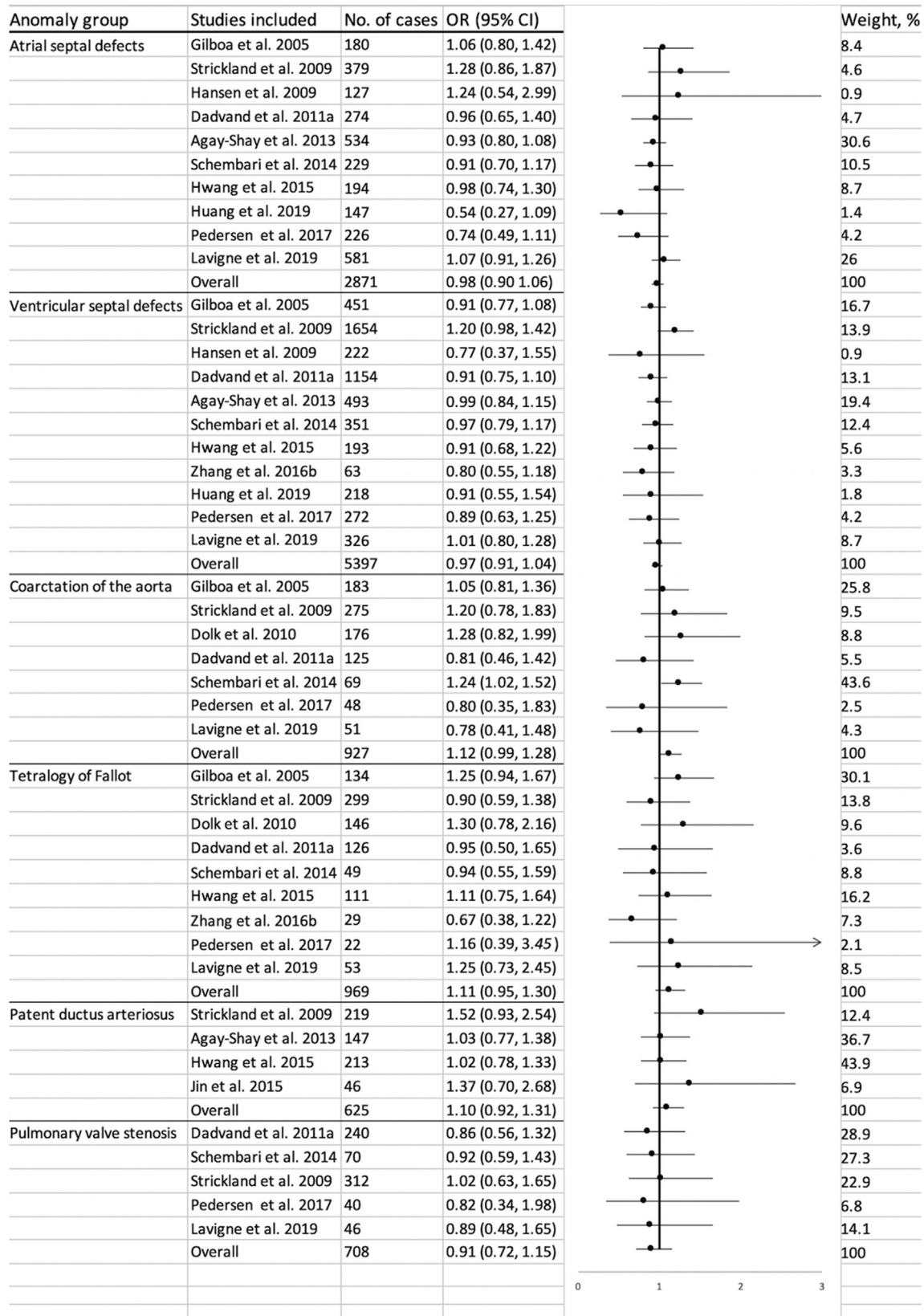


Fig. 6. Summary of meta-analysis of studies for the association between NO₂ exposure and congenital heart defects.

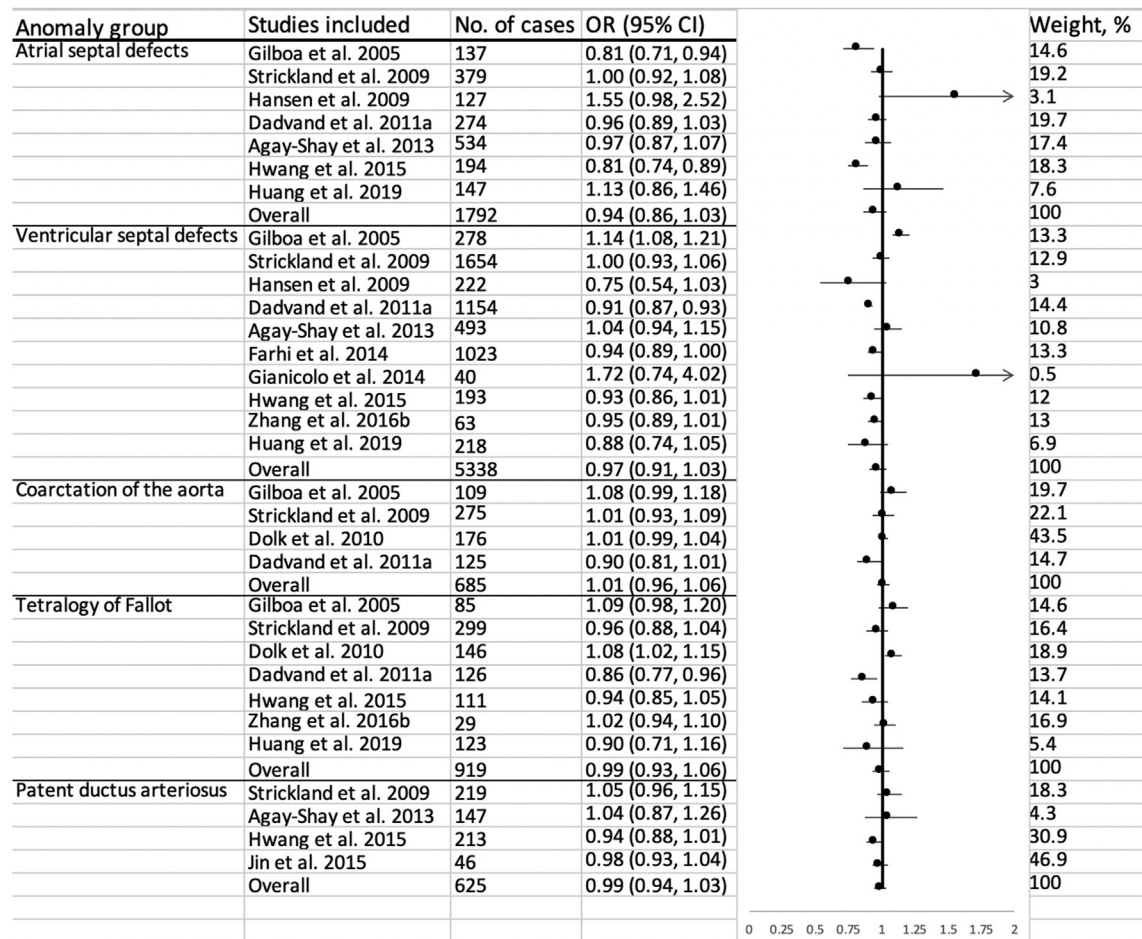


Fig. 7. Summary of meta-analysis of studies for the association between SO_2 exposure and congenital heart defects.

proposed by previous studies (Botto et al., 2007; Jacobs et al., 2007), it's not specific enough to apply these systems for the diagnostics information from routine registries.

The results of this study should be interpreted with caution because some pollution–cardiac defect pairs are based on few studies, and thus were subject to some degree of statistical heterogeneity. For another, the total numbers of cases included in the different pollution–cardiac defect combination meta-analyses were varied, ranging from 375 to >6,000. Significant results for some pollution–cardiac defect pairs were found, but not all the significant associations were robust in the sensitive analyses, and further studies are necessary to confirm our findings. Additionally, we cannot confirm the consistency for the significant results of continuous exposure analyses and high-vs. low-exposure analyses. There is no consensus on which of the analysis approach is most appropriate: using a continuous analysis, which assumes a log-linear association between exposure and outcome, but as previous study reported that linear relationship can't automatically be assumed in the field of congenital anomaly, because selecting survival of more viable fetuses related to the same exposure may lead to nonlinearity (Ritz, 2010). On the other hand, the high vs. low meta-analyses combine categories of very different exposure levels across studies and has lower power due to the small numbers of cases in these more extreme exposure categories (Vrijheid et al., 2011). Future studies could report both the two types of analyses when possible, in order to improve the interpretability as well as comparability with other studies.

The observed heterogeneity across the included studies may arise from inherent differences in the study designs, settings, and populations as well as methodological approaches. Firstly, for the study design, we consider that heterogeneity from the exposure assessment methods may be relatively similar: studies assessed commonly the same exposure windows, same air pollutants, and similar distance-based exposure metrics, but some studies like Dolk et al. (2009) using a purely spatial model and Strickland et al., (2009) using a purely temporal exposure model, which were deviated from other studies, and thus could partly introduce heterogeneity in our analyses. Secondly, with respect to the study setting, population characteristics, exposure levels, pollutant constitutions, and underlying cardiac defect risks were different in the varied areas, this may result in different dose–response relationships. Thirdly, the covariates adjusted in the included studies were different, which could also introduce heterogeneity. As mentioned above, definition and classification of CHDs are different among the included studies, and therefore may contribute to the observed heterogeneity. Because all the selected studies were conducted based on routinely registered data and thus selection and recall biases didn't likely to play a large role in the observed heterogeneity.

The wide range of covariates adjusted in the individual studies included and different residual confounding structures could attribute to the heterogeneity in our meta-analysis. However, there is limited evidence for the contribution of specific factor to CHDs causation with exception of maternal smoking (Baldacci et al.,

2018; Nicoll, 2018), none of the other suspected risk factors, such as advanced maternal age, maternal perinatal diseases, maternal medication use, folate deficiency, alcohol consumption, or low socioeconomic status, are possibly to explain a large proportion of causation. For example, in the study of Ritz et al. (2007), they assessed the relationship between air pollution exposure and preterm birth, and found that after adjusting maternal smoke and environmental tobacco, alcohol consumption, occupation, and income, the odds ratio changed less than 5%. Maternal smoking during pregnancy, for example, has been linked to multiple types of CHDs, but only eight of the included studies adjusted maternal smoking in the analysis models (see Table 1).

The possible teratogenic mechanisms of air pollutants are not fully understood and requires further study, but substantial evidence indicated that placental inflammation and oxidative stress have been involved in the mechanisms, which play important roles in epigenetic changes including alteration of DNA methylation (Baccarelli and Bollati, 2009; Mazzoli-Rocha et al., 2010). Such epigenetic modifications during pregnancy could impair normal embryo development and thus lead to congenital anomalies (Salavati et al., 2018).

It is widely speculated that the etiology of most CHDs may arise from a complex combination of environmental and genetic factors that interfere with cardiogenesis process (Cresci et al., 2011; Krauss and Hong, 2016; Lage et al., 2012). Assessment of gene-environmental interactions that contribute to CHDs development in large-scale studies may be helpful in mechanistic understanding the association of maternal air pollution exposure with CHDs risk in offspring in future studies. Furthermore, specific micro- and macronutrients in the diet have been assumed to modify the effects of toxicants exposure on adverse birth outcomes such as low birth weight, fetal growth restriction, or preterm birth through possible mechanisms of inflammatory and oxidative stress (Kannan et al., 2006; Jedrychowski et al., 2010). The interactions between nutrients and air pollutants are also worth of exploring in the causation of CHDs.

5. Conclusions and recommendations

Our study provided some evidence of an effect of air pollutants on some specific CHDs subtypes risk. Given the ubiquitous exposure of the population to air pollution and the rareness but serious outcomes of CHDs, even small increases in risk may therefore carry a large public health implication. Future studies are needed to confirm our findings and would also benefit from improved exposure assessment, better outcome ascertainment as well as comprehensive control of confounders in this area. Furthermore, increased understanding of underlying mechanisms of teratogenic action of air pollutants is needed through integrating mechanistic biomarkers. Assessment of the interactions between dietary factors and genetic polymorphisms may also provide further insights into mechanisms.

Competing financial interests

The authors have nothing on disclose.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.chemosphere.2020.126668>.

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